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Allenylation

# Theoretical Studies of Allene Synthesis through Cadmium Iodide-Mediated Allenylation of Terminal Alkynes

Yulin Han[a, b] and Xue Zhang*[a, b]

**Abstract:** The cadmium iodide-mediated reaction between an amine and two molecules of terminal alkynes for the synthesis of allenes has been investigated by using DFT calculations. On the basis of these calculations, a plausible mechanism has been proposed. The results also reveal that pyrrolidine serves not only as a reactant but also as a shuttle in a proton relay.

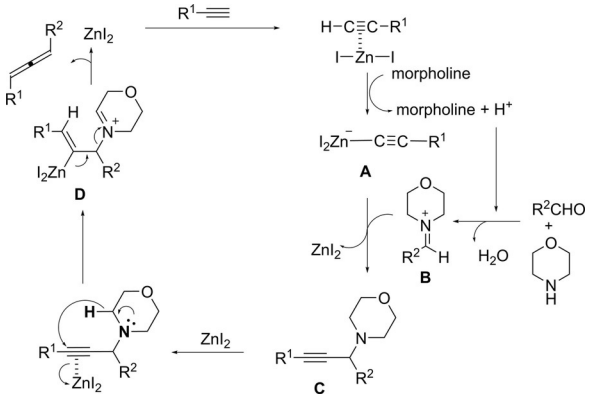
## Introduction

Allenes are an important class of compounds that serve as invaluable synthons in synthetic organic chemistry,[1] and these moieties have been found in many natural products and pharmaceuticals.[1a] In the past, there has been much effort to develop synthetic methods for their preparation.[2,3] Among these approaches, the metal-promoted allenylation of terminal alkynes (ATA) with aldehydes in the presence of amines has emerged as a practical synthetic method for the construction of substituted allenes.[3a,b,4–6] By using ATA reactions, mono-,[3a,b,4] 1,3-di-,[5] and trisubstituted allenes[6] are now easily prepared from terminal alkynes, aldehydes or ketones, and amines.

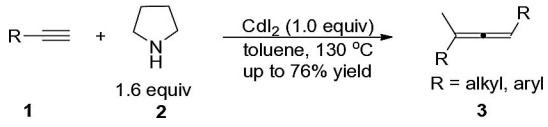
In terms of the mechanism, the formation of propargylamine **C** from the A$^3$-coupling[7,8] of an alkyne, aldehyde or ketone, and amine is one of the key steps (Scheme 1). This proceeds through the reaction of 1-alkynyl zinc species **A** with ketoniminium ion **B**, which is generated in situ from the condensation of the aldehyde with morpholine. After the formation of **C**, a subsequent [1,5] hydride migration and β-elimination[9] produce the final allene product (i.e., a $ZnI_2$-promoted ATA reaction, Scheme 1).[9] This mechanism is supported by our detailed theoretical investigations.[9a]

Recently, a cadmium iodide-mediated tandem reaction that involves an amine and two molecules of terminal alkynes for the synthesis of trisubstituted allenes was developed by our group (Scheme 2).[10] It is worth noting that in contrast to the classical ATA reactions mentioned above, aldehydes or ketones are not necessary for the formation of allene by using this method. In the absence of an aldehyde or ketone, propargylamine **C** could not be formed through the A$^3$-coupling process. Consequently, the above-mentioned mechanism is inapplicable to this protocol, especially to the generation of ketoniminium ion **B**.

With our continued interest in the development of synthetic methods for the preparation of allenes,[4a,c–g,5a–g,6] and a subsequent desire to understand their mode from a mechanistic perspective,[9] we proceeded to perform a computational investigation of this $CdI_2$-mediated ATA reaction that involves an amine and two terminal alkyne molecules.




**Scheme 1.** Proposed mechanism for $ZnI_2$-promoted ATA Reaction.




**Scheme 2.** $CdI_2$-mediated ATA reaction.

[a] Y. Han, Dr. X. Zhang
State Key Laboratory of Organometallic Chemistry
Shanghai Institute of Organic Chemistry
Chinese Academy of Sciences
345 Lingling Lu, Shanghai 200032 (P. R. China)
E-mail: xzhang@sioc.ac.cn

[b] Y. Han, Dr. X. Zhang
University of Chinese Academy of Sciences
Beijing 100049 (P. R. China)

Supporting information and the ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/ajoc.201700389.


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## Results and Discussion

In the absence of an aldehyde or ketone, the ketoniminium ion can be generated by the hydroamination of an alkyne.[11] The calculated free energy profiles for the hydroamination of propyne with pyrrolidine are provided in Figure 1, and the generation of both Markovnikov and anti-Markovnikov products has been investigated in detail. The depicted structures of the stationary points along the schematic of the Markovnikov pathway are illustrated in Figure 1.[12]

The initial formation of the complex **Int1** (selected as a free energy reference) generated by the simultaneous coordination of the pyrrolidine and alkyne to $CdI_2$ leads to a decrease in energy by 20.7 kcal mol$^{-1}$. Complex **Int1** is a tetrahedral four-coordinate $Cd^{II}$ complex. The coordination of $Cd^{II}$ to the C1–C2 triple bond of the alkyne results in the lengthening of the bond from 1.200 Å (for **1a** without $Cd^{II}$ coordination) to 1.215 Å (in **Int1**). Thus, the coordination of $Cd^{II}$ reduces the strength of the triple bond and increases its reactivity towards nucleophilic attack by pyrrolidine. Natural bond orbital (NBO) analysis reveals that the Wiberg bond index of the C1–C2 triple bond in **Int1** is 2.81, which is smaller than that of C1–C2 triple bond in **1a** (2.91). This difference is the result of the bonding π electrons of the triple bond being donated to the empty orbital of $Cd^{II}$ and $Cd^{II}$ backdonating its d electrons into the empty antibonding orbital (π*) of the triple bond. When the subsequent pyrrolidine attack takes place at the C1 atom of the alkyne, the Markovnikov product is produced (solid line in Figure 1). In contrast, the anti-Markovnikov product is provided when the C2 atom participates in the attack (dotted line in Figure 1). The nucleophilic attack that proceeds through the Markovnikov pathway is endergonic ($\Delta G_{sol} = 10.4$ kcal mol$^{-1}$) and requires a free energy barrier of 25.6 kcal mol$^{-1}$ (**TS1_1**), which is lower than **TS1_2** for the anti-Markovnikov attack by 2.4 kcal mol$^{-1}$ (28.0 kcal mol$^{-1}$).

The expected enamine **Int6_1** of the Markovnikov pathway could be formed by a proton migration from the amine N1 atom to the C2 atom of zwitterionic intermediate **Int2_1**. However, the activation energy of this direct proton transfer is computed to be 50.9 kcal mol$^{-1}$ (**TS2′**) with respect to **Int2_1**, which suggests that this step is not kinetically favorable. The ring strain of the four-membered cyclic structure of **TS2′** may lead to its high energy.

A kinetically more favorable pathway for the proton transfer, which is assisted by an additional pyrrolidine molecule, has been discovered.[11d,13] Intermediate **Int3_1** is strongly stabilized (15.6 kcal mol$^{-1}$) by the formation of two hydrogen bonds to an additional pyrrolidine molecule [$d$(N2···H1) = 1.789 Å and $d$(I···H2) = 3.030 Å]. The subsequent abstraction of the H1 proton from the N1 atom and transfer to the N2 atom (in **Int3_1**) requires a free energy barrier of only 1.5 kcal mol$^{-1}$ (**TS2_1**) to afford the pyrrolidinium ion intermediate **Int4_1**. This process is exergonic relative to **Int3_1** by 6.9 kcal mol$^{-1}$. Subsequently, the proton from the pyrrolidinium ion undergoes another migration to the C2 alkyne carbon to produce enamine-bound intermediate **Int5_1**. This step is calculated to be highly exergonic (38.6 kcal mol$^{-1}$ relative to **Int4_1**) and requires a free energy barrier of only 3.9 kcal mol$^{-1}$. Finally, enamine **Int6_1** is released from coordination to the $CdI_2$ to complete this alkyne hydroamination process. Thus, our calcula-

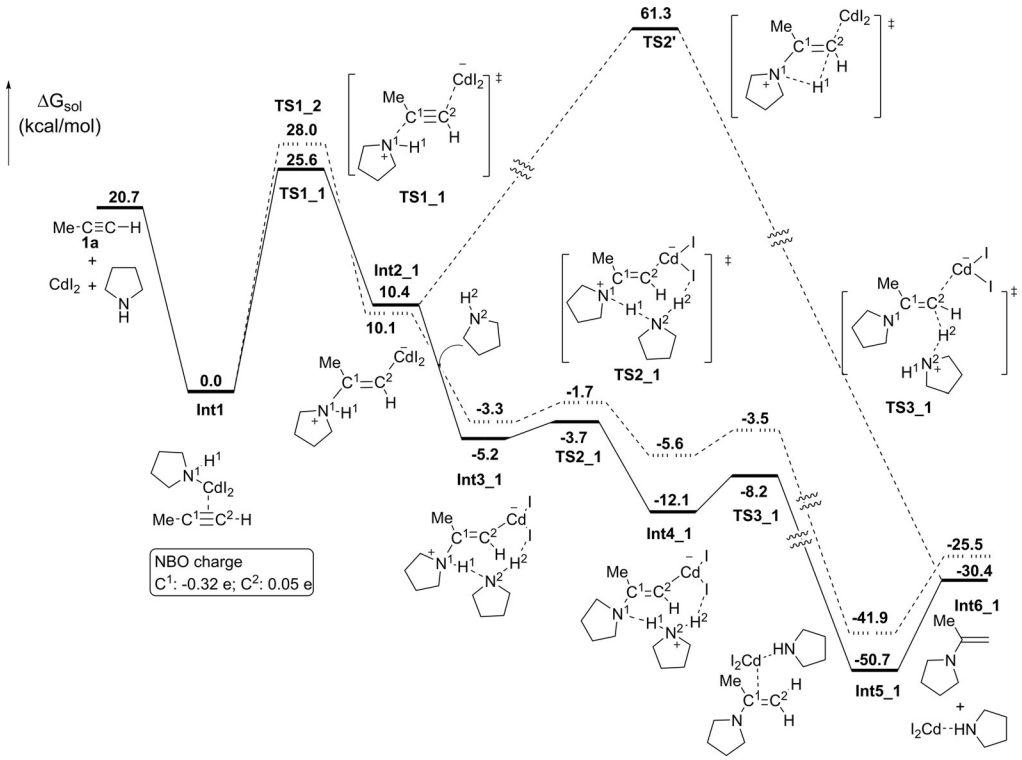



**Figure 1.** Free energy profiles for hydroamination of propyne with pyrrolidine (solid line = Markovnikov pathway, dotted line = anti-Markovnikov pathway).


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tions show that the assistance of an extra pyrrolidine molecule changes the direct proton transfer to a irreversible stepwise protonation–deprotonation process that is highly exergonic by 45.5 kcal mol$^{-1}$ (**Int5_1** relative to **Int3_1**) and requires a total free energy barrier of only 3.9 kcal mol$^{-1}$, which is a significant reduction in the activation energy.

The hydroamination of an alkyne through an anti-Markovnikov pathway is shown for comparision (dotted line in Figure 1, for details see Supporting Information, Figure S1). The computed potential energy surface is qualitatively similar to that of a Markovnikov process, but this pathway is more energy demanding, especially the step for the nucleophilic attack (28.0 kcal mol$^{-1}$ for **TS1_2** vs. 25.6 kcal mol$^{-1}$ for **TS_1**). Thus, Markovnikov intermediate **Int2_1** is generated overwhelmingly. It is worth noting that the coordination of $CdI_2$ to the alkyne triple bond of **Int1** is unsymmetrical with a longer Cd···C1 (2.995 Å) than Cd···C2 distance (2.692 Å), which indicates that the terminal C2 carbon is more strongly coordinated to the cadmium center than the substituted C1 carbon. Moreover, natural population analysis (NPA) reveals that the natural atomic charge on the C2 atom of **Int1** is much more negative than that on the C1 atom (−0.32 vs. 0.05 e), and, thus, the C2 carbon is less reactive than C1 atom towards nucleophilic attack by pyrrolidine. Our investigation accounts for the selectivity of the alkyne hydroamination through the Markovnikov pathway, which is also in accordance with the experimental observations of the Markovnikov product as the sole product.[10]

Propargylamine is one of the key intermediates in the synthesis of allenes, and this has been verified experimentally by using this nonclassical ATA process.[10] According to our understanding of ATA reactions,[9a] propargylamine may be produced by the electrophilic attack on the 1-alkynyl cadmium species by the ketoniminium ion. In this process, the ketoniminium ion is probably generated by enamine protonation. Figure 2 presents the free energy profile for the generation of propargylamine from the initial complex **Int1** (free energy reference). The process starts with the isomerization of **Int1** into **Int7**, in which two hydrogen bonds are formed between the pyrrolidine molecule and the alkyne-coordinated $CdI_2$ [$d$(N1···H) = 1.921 Å and $d$(I···H1) = 3.071 Å], and leads to an energy increase of 14.9 kcal mol$^{-1}$. The C2–H single bond is weakened by the formation of hydrogen bonds, which results in an increase in the C2–H bond distance by 0.040 Å (1.109 Å in **Int7** vs. 1.069 Å in **Int1**). The subsequent abstraction of the proton from the C2 carbon of the terminal alkyne and transfer to the N1 atom of pyrrolidine (in **Int7**, Figure 2) requires an activation energy of 4.9 kcal mol$^{-1}$ (**TS4**) to afford complex **Int8**, which contains hydrogen-bonding interactions between the 1-alkynyl cadmium species and the pyrrolidinium ion.

The in situ generated pyrrolidinium ion may then act as a proton source for the conversion of the enamine intermediate into a ketoniminium ion. This enamine protonation step is calculated to require an activation energy of only 6.0 kcal mol$^{-1}$ through **TS5** to afford intermediate **Int10**, which is a hydrogen-bonded complex of the ketoniminium ion, pyrrolidine, and propyn-1-yl cadmium species. We tried to evaluate the possibility of the direct proton transfer from the terminal alkyne to the enamine, but we were unable to locate the transition-state structures of this transformation with or without the assistance of $CdI_2$. Obviously, the two precursors for the propargylamine formation, the propyn-1-yl cadmium species and ketoniminium ion, could be easily generated with the assistance of a pyrrolidine molecule. The pyrrolidine may act as a proton shuttle once again by abstracting a proton from the terminal alkyne carbon atom and subsequently protonating the enamine to give a ketoniminium ion.

The reaction continued with the release of pyrrolidine from **Int10** to yield the anion/cation complex **Int11**, which is endergonic by 7.5 kcal mol$^{-1}$ relative to **Int10**. The subsequent electrophilic attack of the ketoniminium ion on the C2 atom attached to cadium requires an energy barrier 10.5 kcal mol$^{-1}$ (**TS6**) to provide propargylamine-$CdI_2$ complex **Int12**, which is lower in energy by 1.8 kcal mol$^{-1}$ relative to **Int1**. The generation of propargylamine from **Int1** can proceed smoothly with the highest free energy barrier of 19.8 kcal mol$^{-1}$ (**TS4**). The purpose of the $CdI_2$ in this process is to activate the terminal alkyne towards electrophilic attack by the iminium ion.

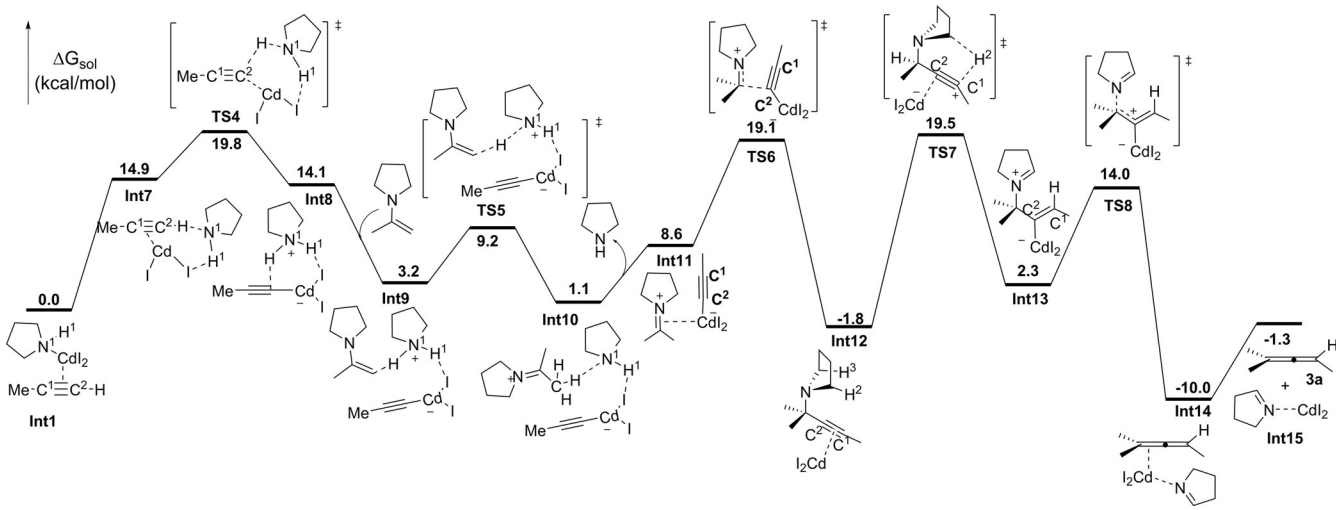



**Figure 2.** Free energy profile for formation of allene **3a** from **Int1**.


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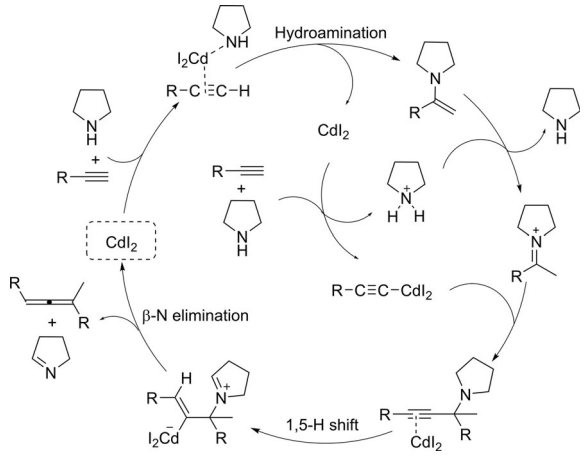
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Allene can be obtained from the propargylamine through a $CdI_2$-promoted [1,5] hydride migration and β-elimination step. The pyrrolidine moiety of **Int12**, with its envelope conformation, has two axial hydrogen atoms (H2 and H3) that are perfectly oriented for migration to the C1 atom. The six-membered cyclic transition-state **TS7** for the [1,5] hydride shift has been located, and during this transformation a $\sigma$ bond between the $Cd^{II}$ and C2 atom is formed to generate intermediate **Int13**. This [1,5] hydride migration step is computed to be slightly endergonic ($\Delta G_{sol} = 4.1$ kcal mol$^{-1}$) and requires an activation barrier of 21.3 kcal mol$^{-1}$ (relative to **Int12**). The coordination with $CdI_2$ enhances the reactivity of the propargylamine by increasing the C1–C2 triple bond distance from 1.211 Å (without $CdI_2$ coordination) to 1.222 Å (in **Int12**). This increase is the result of electron donation and back-donation between the triple bond and cadmium(II) ion. Thus, the coordination of $CdI_2$ to the propargylamine in **Int12** facilitates the subsequent [1,5] hydride migration and β-elimination step. The β-elimination to give intermediate **Int14** is relatively easier with an activation barrier of only 11.7 kcal mol$^{-1}$ through **TS8** (Figure 2). The simultaneous breaking of the C–N and C–Cd bonds irreversibly provide the product **Int14**, which is 12.3 kcal mol$^{-1}$ lower in energy than **Int13**. The final step of the process involves the release of **3a** from its coordination to cadmium. The release of allene is calculated to be endergonic by 8.7 kcal mol$^{-1}$ to produce the $CdI_2$-coordinated 3,4-dihydro-2*H*-pyrrole **Int15** and the final allene product **3a**. Thus, during the formation of allene from propargylamine, the [1,5] hydride migration step is computed to have the highest relative energy barrier of 21.3 kcal mol$^{-1}$ (**TS7** relative to **Int12**). The key role of the $CdI_2$ in this process is to activate the alkyne towards hydride migration.

It should be noted that the highest relative energy barrier (21.3 kcal mol$^{-1}$, **TS7** in Figure 2) for the formation of allene by the **Int1** process (Figure 2) is lower than that of the process for the formation of the enamine (25.6 kcal mol$^{-1}$, **TS1_1** in Figure 1). If we only consider the energy barriers, it seems that enamine formation should never take place. Nevertheless, the formation of the allene requires the enamine as a reactant to form the thermodynamically more stable intermediates.

## Conclusions

In summary, the mechanism of this $CdI_2$-mediated ATA reaction in the presence of an amine and two terminal alkyne molecules was investigated in detail. On the basis of the DFT calculations, a modified reaction mechanism has been proposed as shown in Scheme 3. First, the $CdI_2$-promoted hydroamination of the alkyne by pyrrolidine produces the enamine according to Markovnikov's rule. The corresponding ketoniminium ion is then obtained by protonation of the enamine. The ketoniminium ion undergoes a reaction with the 1-alkynyl cadmium species, obtained from a terminal alkyne in the presence of pyrrolidine, to provide the propargylamine. Finally, [1,5] hydride migration and β-elimination in the presence of $CdI_2$ provides the desired trisubstituted allene. Because the hydroamination of the alkyne to form the enamine features the highest relative




**Scheme 3.** The modified reaction mechanism.

energy barrier (25.6 kcal mol$^{-1}$, **TS1_1**) of the entire process, this is the rate-determining step. In particular, our investigation suggests that pyrrolidine not only act as a reactant but also as a shuttle in the proton relay process that occurs during the hydroamination of the alkyne and the formation of the propyn-1-yl cadmium species and ketoniminium ion. Moreover, our calculations reveal that the main role of $CdI_2$ is to facilitate the hydroamination reaction by activating the terminal alkyne and accelerate the [1,5] hydride migration by activating the propargylamine.

## Experimental Section

**Computational Methods**: All calculations were carried out by using the Gaussian 09[14] package. Geometries were fully optimized by using the density functional theory B3LYP method[15]. The LANL2DZ basis set[16] with effective core potential (ECP) was used for the Cd and I atoms, and the 6-31G (d)[16b,17] basis set was used for the other atoms. Harmonic vibration frequency calculations were conducted at the same level of theory to verify the stationary points as minima (no imaginary frequency) or saddle points (one imaginary frequency). Intrinsic reaction coordinate (IRC)[18] calculations were performed to confirm the connection of the transition-state structures with their corresponding reactants and products. The single point energies and solvent effects in toluene were computed by M06[19]/SDD-6-311+G(d,p) basis sets using the SMD[20] solvation model. All of the energies discussed in this paper are Gibbs free energies at 298 K ($\Delta G_{sol}$) in toluene [dielectric constant ($\varepsilon$) = 2.37], unless otherwise specified. Fully optimized structures are illustrated using CYLView. We utilized propyne (**1a**, R = Me in Scheme 2) as the working model of the terminal alkyne to reduce the computational cost.

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## Conflict of interest

The authors declare no conflict of interest.

**Keywords:** alkynes · allenes · cadmium · density functional calculations · hydroamination

[1] For recent reviews on the chemistry of allenes, see: a) A. Hoffmann-Röder, N. Krause, *Angew. Chem. Int. Ed.* **2004**, *43*, 1196; *Angew. Chem.* **2004**, *116*, 1216; b) S. Ma, *Chem. Rev.* **2005**, *105*, 2829; c) S. Ma, *Aldrichimica Acta* **2007**, *40*, 91; d) M. Brasholz, H. U. Reissig, R. Zimmer, *Acc. Chem. Res.* **2009**, *42*, 45; e) S. Ma, *Acc. Chem. Res.* **2009**, *42*, 1679; f) B. Alcaide, P. Almendros, T. M. del Campo, *Chem. Eur. J.* **2010**, *16*, 5836; g) C. Aubert, L. Fensterbank, P. Garcia, M. Malacria, A. Simonneau, *Chem. Rev.* **2011**, *111*, 1954; h) N. Krause, C. Winter, *Chem. Rev.* **2011**, *111*, 1994; i) F. Inagaki, S. Kitagaki, C. Mukai, *Synlett* **2011**, 594; j) F. López, J. L. Mascareñas, *Chem. Eur. J.* **2011**, *17*, 418; k) S. Yu, S. Ma, *Angew. Chem. Int. Ed.* **2012**, *51*, 3074; *Angew. Chem.* **2012**, *124*, 3128; l) J. Ye, S. Ma, *Acc. Chem. Res.* **2014**, *47*, 989; m) S. Kitagaki, F. Inagaki, C. Mukai, *Chem. Soc. Rev.* **2014**, *43*, 2956; n) F. López, J. L. Mascareñas, *Chem. Soc. Rev.* **2014**, *43*, 2904; o) J. Le, J. Muzart, *Chem. Soc. Rev.* **2014**, *43*, 3003; p) J. M. Alonso, M. T. Quirós, M. P. Muñoz, *Org. Chem. Front.* **2016**, *3*, 1186.

[2] For selected reviews on the synthesis of allenes, see: a) L. K. Sydnes, *Chem. Rev.* **2003**, *103*, 1133 – 1150; b) N. Krause, A. S. K. Hashmi (Eds.), *Modern Allene Chemistry, Vols. 1 and 2*, Wiley-VCH, Weinheim, **2004**; c) N. Krause, A. Hoffmann-Röder, *Tetrahedron* **2004**, *60*, 11671; d) G. B. Hammond, *ACS Symp. Ser.* **2005**, *911*, 204; e) K. M. Brummond, J. E. Deforrest, *Synthesis* **2007**, 795; f) M. Ogasawara, *Tetrahedron: Asymmetry* **2009**, *20*, 259; g) S. Yu, S. Ma, *Chem. Commun.* **2011**, *47*, 5384; h) J. Ye, S. Ma, *Org. Chem. Front.* **2014**, *1*, 1210; i) R. K. Neff, D. E. Frantz, *ACS Catal.* **2014**, *4*, 519.

[3] For selected examples of the synthesis of allenes, see: a) P. Crabbé, H. Fillion, D. André, J. L. Luche, *J. Chem. Soc., Chem. Commun.* **1979**, 859; b) H. Nakamura, T. Sugiishi, Y. Tanaka, *Tetrahedron Lett.* **2008**, *49*, 7230; c) V. K. Y. Lo, M.-K. Wong, C.-M. Che, *Org. Lett.* **2008**, *10*, 517; d) V. K.-Y. Lo, C.-Y. Zhou, M.-K. Wong, C.-M. Che, *Chem. Commun.* **2010**, *46*, 213; e) H. Nakamura, M. Ishikura, T. Sugiishi, T. Kamakura, J. F. Biellmann, *Org. Biomol. Chem.* **2008**, *6*, 1471; f) A. G. Myers, B. Zheng, *J. Am. Chem. Soc.* **1996**, *118*, 4492; g) Q. Xiao, Y. Xia, H. Li, Y. Zhang, J. Wang, *Angew. Chem. Int. Ed.* **2011**, *50*, 1114; *Angew. Chem.* **2011**, *123*, 1146; h) X. Zhao, Z. Zhong, P. Peng, W. Zhang, J. Wang, *Chem. Commun.* **2009**, 2535.

[4] a) S. Ma, H. Hou, S. Zhao, G. Wang, *Synthesis* **2002**, 1643; b) U. Kazmaier, S. Lucas, M. Klein, *J. Org. Chem.* **2006**, *71*, 2429; c) J. Kuang, S. Ma, *J. Org. Chem.* **2009**, *74*, 1763; d) B. Chen, N. Wang, W. Fan, S. Ma, *Org. Biomol. Chem.* **2012**, *10*, 8465; e) H. Luo, S. Ma, *Eur. J. Org. Chem.* **2013**, 3041; f) J. Kuang, X. Xie, S. Ma, *Synthesis* **2013**, *45*, 592; g) X. Huang, C. Fu, S. Ma, *Synthesis* **2014**, *46*, 2917.

[5] a) J. Kuang, S. Ma, *J. Am. Chem. Soc.* **2010**, *132*, 1786; b) J. Kuang, H. Luo, S. Ma, *Adv. Synth. Catal.* **2012**, *354*, 933; c) X. Huang, T. Cao, Y. Han, X. Jiang, W. Lin, J. Zhang, S. Ma, *Chem. Commun.* **2015**, *51*, 6956; d) X. Tang, X. Huang, T. Cao, Y. Han, X. Jiang, W. Lin, Y. Tang, J. Zhang, Q. Yu, C. Fu, S. Ma, *Org. Chem. Front.* **2015**, *2*, 688; e) J. Ye, S. Li, B. Chen, W. Fan, J. Kuang, J. Liu, Y. Liu, B. Miao, B. Wan, Y. Wang, X. Xie, Q. Yu, W. Yuan, S. Ma, *Org. Lett.* **2012**, *14*, 1346; f) J. Ye, W. Fan, S. Ma, *Chem. Eur. J.* **2013**, *19*, 716; g) R. Lü, J. Ye, T. Cao, B. Chen, W. Fan, W. Lin, J. Liu, H. Luo, B. Miao, S. Ni, X. Tang, N. Wang, Y. Wang, X. Xie, Q. Yu, W. Yuan, W. Zhang, C. Zhu, S. Ma, *Org. Lett.* **2013**, *15*, 2254; h) S. Kitagaki, M. Komizu, C. Mukai, *Synlett* **2011**, *8*, 1129; i) M. Periasamy, N. Sanjeevakumar, M. Dalai, R. Gurubrahamam, P. O. Reddy, *Org. Lett.* **2012**, *14*, 2932.

[6] X. Tang, C. Zhu, T. Cao, J. Kuang, W. Lin, S. Ni, J. Zhang, S. Ma, *Nat. Commun.* **2013**, *4*, 2450.

[7] For reviews of the $A^3$-coupling reaction, see: a) B. M. Trost, A. H. Weiss, *Adv. Synth. Catal.* **2009**, *351*, 963; b) L. Zani, C. Bolm, *Chem. Commun.* **2006**, 4263; c) C. Wei, Z. Li, C.-J. Li, *Synlett* **2004**, 1472; d) W.-J. Yoo, L. Zhao, C.-J. Li, *Aldrichimica Acta* **2011**, *44*, 43; e) G. Blay, A. Monleón, J. R. Pedro, *Curr. Org. Chem.* **2009**, *13*, 1498; f) C.-J. Li, *Acc. Chem. Res.* **2009**, *42*, 335; g) V. A. Peshkov, O. P. Pereshivko, E. V. Van der Eycken, *Chem. Soc. Rev.* **2012**, *41*, 3790.

[8] For selected examples of the $A^3$-coupling reaction, see: a) J. B. Bariwal, D. S. Ermolat'ev, E. V. Van der Eychen, *Chem. Eur. J.* **2010**, *16*, 3281; b) W.-J. Yoo, C.-J. Li, *Adv. Synth. Catal.* **2008**, *350*, 1503; c) S. B. Park, H. Alper, *Chem. Commun.* **2005**, 1315; d) C. Wei, C.-J. Li, *J. Am. Chem. Soc.* **2003**, *125*, 9584; e) C.-J. Li, C. Wei, *Chem. Commun.* **2002**, 268; f) S. Sakaguchi, T. Kubo, Y. Ishii, *Angew. Chem. Int. Ed.* **2001**, *40*, 2534; *Angew. Chem.* **2001**, *113*, 2602; g) N. Gommermann, C. Koradin, K. Polborn, P. Knochel, *Angew. Chem. Int. Ed.* **2003**, *42*, 5763; *Angew. Chem.* **2003**, *115*, 5941; h) N. Gommermann, P. Knochel, *Chem. Commun.* **2004**, 2324; i) N. Gommermann, P. Knochel, *Chem. Commun.* **2005**, 4175; j) N. Gommermann, P. Knochel, *Chem. Eur. J.* **2006**, *12*, 4380; k) P. Aschwanden, C. R. J. Stephenson, E. M. Carreira, *Org. Lett.* **2006**, *8*, 2437; l) V. K.-Y. Lo, Y. Liu, M.-K. Wong, C.-M. Che, *Org. Lett.* **2006**, *8*, 1529; m) L. Rubio-Pérez, M. Iglesias, J. Munárriz, V. Polo, J. J. Pérez-Torrente, L. A. Oro, *Chem. Eur. J.* **2015**, *21*, 17701.

[9] a) X. Zhang, *Asian J. Org. Chem.* **2014**, *3*, 309; b) C. Liao, B. Li, J. Wang, Y. Wang, *Chin. J. Chem.* **2012**, *30*, 951; c) M. González, R. A. Rodríguez, M. M. Cid, C. S. Lopez, *J. Comput. Chem.* **2012**, *33*, 1236.

[10] X. Tang, Y. Han, S. Ma, *Org. Lett.* **2015**, *17*, 1176.

[11] a) G. Ciancaleoni, L. Belpassi, D. Zuccaccia, F. Tarantelli, P. Belanzoni, *ACS Catal.* **2015**, *5*, 803; b) G. Ciancaleoni, S. Rampino, D. Zuccaccia, F. Tarantelli, P. Belanzoni, L. Belpassi, *J. Chem. Theory Comput.* **2014**, *10*, 1021; c) G. Mazzone, N. Russo, E. Sicilia, *J. Chem. Theory Comput.* **2015**, *11*, 581; d) M. Katari, M. N. Rao, G. Rajaraman, P. Ghosh, *Inorg. Chem.* **2012**, *51*, 5593.

[12] The results for the anti-Markovnikov pathway are provided in the Supporting Information.

[13] For examples of DFT calculations that demonstrate the proton transfer assisted by a base or anion, see: a) F.-Q. Shi, X. Li, Y. Xia, L. Zhang, Z.-X. Yu, *J. Am. Chem. Soc.* **2007**, *129*, 15503; b) A. Rossin, L. Gonsalvi, A. D. Phillips, O. Maresca, A. Lledós, M. Peruzzini, *Organometallics* **2007**, *26*, 3289; c) Y. Xia, Y. Liang, Y. Chen, M. Wang, L. Jiao, F. Huang, S. Liu, Y. Li, Z.-X. Yu, *J. Am. Chem. Soc.* **2007**, *129*, 3470; d) Y. Xia, A. S. Dudnik, Y. Li, V. Gevorgyan, *Org. Lett.* **2010**, *12*, 5538; e) Y. Qiu, C. Fu, X. Zhang, S. Ma, *Chem. Eur. J.* **2014**, *20*, 10314.

[14] Gaussian 09, Revision A.02, M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G. A. Petersson, H. Nakatsuji, M. Caricato, X. Li, H. P. Hratchian, A. F. Izmaylov, J. Bloino, G. Zheng, J. L. Sonnenberg, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, J. A. Montgomery, Jr., J. E. Peralta, F. Ogliaro, M. Bearpark, J. J. Heyd, E. Brothers, K. N. Kudin, V. N. Staroverov, R. Kobayashi, J. Normand, K. Raghavachari, A. Rendell, J. C. Burant, S. S. Iyengar, J. Tomasi, M. Cossi, N. Rega, J. M. Millam, M. Klene, J. E. Knox, J. B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R. E. Stratmann, O. Yazyev, A. J. Austin, R. Cammi, C. Pomelli, J. W. Ochterski, R. L. Martin, K. Morokuma, V. G. Zakrzewski, G. A. Voth, P. Salvador, J. J. Dannenberg, S. Dapprich, A. D. Daniels, O. Farkas, J. B. Foresman, J. V. Ortiz, J. Cioslowski, D. J. Fox, Gaussian, Inc., Wallingford CT, **2009**.

[15] P. C. Hariharan, J. A. Pople, *Theor. Chim. Acta* **1973**, *28*, 213.

[16] a) K. Fukui, *Acc. Chem. Res.* **1981**, *14*, 363; b) P. J. Hay, W. R. Wadt, *J. Chem. Phys.* **1985**, *82*, 299; c) T. H. Dunning, Jr., P. J. Hay in *Modern Theoretical Chemistry* (Ed.: H. F. Schaefer, III), Plenum Press, New York, **1977**, pp. 1 – 28.

[17] a) P. J. Hay, W. R. Wadt, *J. Chem. Phys.* **1985**, *82*, 270; b) W. R. Wadt, P. J. Hay, *J. Chem. Phys.* **1985**, *82*, 284.

[18] a) K. Fukui, *J. Phys. Chem.* **1970**, *74*, 4161; b) C. Gonzalez, H. B. Schlegel, *J. Chem. Phys.* **1989**, *90*, 2154; c) C. Gonzalez, H. B. Schlegel, *J. Phys. Chem.* **1990**, *94*, 5523.

[19] Y. Zhao, D. G. Truhlar, *Theor. Chem. Acc.* **2008**, *120*, 215.

[20] A. V. Marenich, C. J. Cramer, D. G. Truhlar, *J. Phys. Chem. B* **2009**, *113*, 6378.

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